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Diaglycon MR Gliclazide

1. NAME OF THE MEDICINAL PRODUCT

Diaglycon MR 30mg Modified Release Tablets
Diaglycon MR 60mg Modified Release Tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Diaglycon MR 30mg Modified Release Tablets: Each tablet contains 30mg of gliclazide
Diaglycon MR 60mg Modified Release Tablets: Each tablet contains 60mg of gliclazide

For the full list of excipients, see section List of excipients.

3. PHARMACEUTICAL FORM

Modified release tablet.

Diaglycon MR 30mg Modified Release Tablets: White to off white, round, uncoated tablet with 'N' on one side and '12' on the other side.
Diaglycon MR 60mg Modified Release Tablets: White to off white, pear shaped, uncoated tablet with 'N' on one side and '11' on the other side.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Non insulin-dependent diabetes (type 2) in adults when dietary measures, physical exercise and weight loss alone are not sufficient to control blood glucose.

4.2 Posology and method of administration

Posology

The daily dose of Diaglycon MR may vary from 30mg to 120mg taken orally in a single intake at breakfast time.

It is recommended to swallow the dose without crushing or chewing.

If a dose is forgotten, there must be no increase in the dose taken the next day.

As with any hypoglycaemic agent, the dose should be adjusted according to the individual patient's metabolic response (blood glucose, HbA_{1c}).

Initial dose

The recommended starting dose is 30mg daily.

If blood glucose is effectively controlled, this dose may be used for maintenance treatment. If blood glucose is not adequately controlled, the dose may be increased to 60, 90 or 120mg daily, in successive steps. The interval between each dose increment should be at least 1 month except in patients whose blood glucose has not reduced after two weeks of treatment. In such cases, the dose may be increased at the end of the second week of treatment.

The maximum recommended daily dose is 120mg.

Switching from Gliclazide 80 mg tablets to Diaglycon MR

One tablet of Gliclazide 80mg is comparable to 30 mg of the modified release formulation. Consequently, the switch can be performed with careful blood monitoring.

Switching from another oral antidiabetic agent to Diaglycon MR

Diaglycon MR can be used to replace other oral antidiabetic agents. The dosage and the half-life of the previous antidiabetic agent should be taken into account when switching to Diaglycon MR.

A transitional period is not generally necessary. A starting dose of 30mg should be used and this should be adjusted to suit the patient's blood glucose response, as described above.

When switching from a hypoglycaemic sulfonylureas with a prolonged half-life, a treatment free period of a few days may be necessary to avoid an additive effect of the two products, which might cause hypoglycaemia. The procedure described for initiating treatment should also be used when switching to treatment with Diaglycon MR 60mg, i.e. a starting dose of 30mg/day, followed by a stepwise increase in dose, depending on the metabolic response.

Combination treatment with other antidiabetic agents

Diaglycon MR can be given in combination with biguanides, alpha glucosidase inhibitors or insulin. In patients not adequately controlled with Diaglycon MR, concomitant insulin therapy can be initiated under close medical supervision.

Special Populations

Elderly

Diaglycon MR should be prescribed using the same dosing regimen recommended for patients under 65 years of age.

Renal impairment

In patients with mild to moderate renal insufficiency, the same dosing regimen can be used as in patients with normal renal function with careful patient monitoring.

Patients at risk of hypoglycaemia

- Undernourished or malnourished,
- Severe or poorly compensated endocrine disorders (hypopituitarism, hypothyroidism, adrenocorticotrophic insufficiency),
- Withdrawal of prolonged and/or high dose corticosteroid therapy,
- Severe vascular disease (severe coronary heart disease, severe carotid impairment, diffuse vascular disease);

It is recommended that the minimum daily starting dose of 30mg is used.

Paediatric population

The safety and efficacy of Diaglycon MR in children and adolescents have not been established. No data are available.

Note: Diaglycon MR 60mg cannot be divided into two/ cut by half.

4.3 Contraindications

This medicine is contra-indicated in case of:

- Hypersensitivity to gliclazide or to any of the excipients as listed under section List of Excipients, other sulfonylurea, sulfonamides,
- Type 1 diabetes,
- Diabetic pre-coma and coma, diabetic keto-acidosis,
- Severe renal or hepatic insufficiency: in these cases the use of insulin is recommended,
- Treatment with miconazole (see section Interaction with other medicinal products and other forms of interaction),
- Lactation (see section Fertility, pregnancy and lactation).

4.4 Special warnings and precautions for use

Hypoglycaemia:

This treatment should be prescribed only if the patient is likely to have a regular food intake (including breakfast). It is important to have a regular carbohydrate intake due to the increased risk of hypoglycaemia if a meal is taken late, if an inadequate amount of food is consumed or if the food is low in carbohydrate. Hypoglycaemia is more likely to occur during low-calorie diets, following prolonged or strenuous exercise, alcohol intake or if a combination of hypoglycaemic agents is being used. Hypoglycaemia may occur following administration of sulfonylurea (see section Undesirable effects). Some cases may be severe and prolonged. Hospitalisation may be necessary and glucose administration may need to be continued for several days.

Careful selection of patients, of the dose used, and clear patient directions are necessary to reduce the risk of hypoglycaemic episodes.

Factors which increase the risk of hypoglycaemia:

- Patient refuses or (particularly in elderly subjects) is unable to co-operate,
- Malnutrition, irregular mealtimes, skipping meals, periods of fasting or dietary changes,
- Imbalance between physical exercise and carbohydrate intake,
- Renal insufficiency,
- Severe hepatic insufficiency,
- Overdose of Diaglycon MR,
- Certain endocrine disorders: thyroid disorders, hypopituitarism and adrenalinsufficiency,
- Certain endocrine disorders: thyroid disorders, hypopituitarism and adrenalinsufficiency,
- Concomitant administration of certain other medicinal products (see section Interaction with other medicinal products and other forms of interaction).

Renal and hepatic insufficiency: the pharmacokinetics and/or pharmacodynamics of gliclazide may be altered in patients with hepatic insufficiency or severe renal failure. A hypoglycaemic episode occurring in these patients may be prolonged, so appropriate management should be initiated.

Patient information:

The risks of hypoglycaemia, together with its symptoms (see section Undesirable effects), treatment, and conditions that predispose to its development, should be explained to the patient and to family members. The patient should be informed of

the importance of following dietary advice, of taking regular exercise, and of regular monitoring of blood glucose levels.

Poor blood glucose control:

Blood glucose control in a patient receiving antidiabetic treatment may be affected by any of the following: St John's Wort (Hypericum Perforatum) preparations (see section Interaction with other medicinal products and other forms of interaction), fever, trauma, infection or surgical intervention. In some cases, it may be necessary to administer insulin.

The hypoglycaemic efficacy of any oral antidiabetic agent, including gliclazide, is attenuated over time in many patients: this may be due to progression in the severity of the diabetes, or to a reduced response to treatment. This phenomenon is known as secondary failure which is distinct from primary failure, when an active substance is ineffective as first-line treatment. Adequate dose adjustment and dietary compliance should be considered before classifying the patient as secondary failure.

Dysglycaemia:

Disturbances in blood glucose, including hypoglycaemia and hyperglycaemia have been reported, in diabetic patients receiving concomitant treatment with fluoroquinolones, especially in elderly patients. Indeed, careful monitoring of blood glucose is recommended in all patients receiving at the same time Diaglycon MR and a fluoroquinolone.

Laboratory tests:

Measurement of glycated haemoglobin levels (or fasting venous plasma glucose) is recommended in assessing blood glucose control. Blood glucose self-monitoring may also be useful.

Treatment of patients with G6PD-deficiency with sulfonylurea agents can lead to haemolytic anaemia. Since gliclazide belongs to the chemical class of sulfonylurea drugs, caution should be used in patients with G6PD-deficiency and a non-sulfonylurea alternative should be considered.

Porphyric patients:

Cases of acute porphyria have been described with some other sulfonylurea drugs, in patients who have porphyria.

Excipients:

Diaglycon MR should not be administered to patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption.

4.5 Interactions with other medicinal products and other forms of interaction

The following products are likely to increase the risk of hypoglycaemia

Contra-indicated combination

- Miconazole** (systemic route, oromucosal gel): increases the hypoglycaemic effect with possible onset of hypoglycaemic symptoms, or even coma.

Combinations which are not recommended

- Phenylbutazone** (systemic route): increases the hypoglycaemic effect of sulfonylurea (displaces their binding to plasma proteins and/or reduces their elimination).

It is preferable to use a different anti-inflammatory agent, or else to warn the patient and emphasise the importance of self-monitoring. Where necessary, adjust the dose during and after treatment with the anti-inflammatory agent.

- Alcohol:** increases the hypoglycaemic reaction (by inhibiting compensatory reactions) that can lead to the onset of hypoglycaemic coma.

Alcohol or medicinal products containing alcohol should be avoided.

Combinations requiring precautions for use

Potentiation of the blood glucose lowering effect and thus, in some instances, hypoglycaemia may occur when one of the following drugs is taken:
•Other antidiabetic agents (insulins, acarbose, metformin, thiazolidinediones, dipeptidyl peptidase-4 inhibitors, GLP-1 receptor agonists), beta-blockers, fluconazole, angiotensin converting enzyme inhibitors (captopril, enalapril), H₂-receptor antagonists, MAOIs, sulfonamides, clarithromycin and nonsteroidal anti-inflammatory agents.

The following products may cause an increase in blood glucose levels

Combination which is not recommended

- Danazol:** diabetogenic effect of danazol.
If the use of this active substance cannot be avoided, warn the patient and emphasise the importance of urine and blood glucose monitoring. It may be necessary to adjust the dose of the antidiabetic agent during and after treatment with danazol.

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Combinations requiring precautions during use

•**Chlorpromazine** (neuroleptic agent): high doses (>100 mg per day of chlorpromazine) increase blood glucose levels (reduced insulin release). Warn the patient and emphasise the importance of blood glucose monitoring. It may be necessary to adjust the dose of the antidiabetic active substance during and after treatment with the neuroleptic agent.

•**Glucocorticoids** (systemic and local route: intra-articular, cutaneous and rectal preparations) and tetracosactrin: increase in blood glucose levels with possible ketosis (reduced tolerance to carbohydrates due to glucocorticoids). Warn the patient and emphasise the importance of blood glucose monitoring, particularly at the start of treatment. It may be necessary to adjust the dose of the antidiabetic active substance during and after treatment with glucocorticoids.

•**Ritodrine, salbutamol, terbutaline: (I.V.)**

Increased blood glucose levels due to beta-2 agonist effects. Emphasise the importance of monitoring blood glucose levels. If necessary, switch to insulin.

•**Saint John's Wort (Hypericum perforatum) preparations:**

Gliclazide exposure is decreased by Saint John's Wort- Hypericum perforatum. Emphasize the importance of blood glucose levels monitoring.

The following products may cause dysglycaemia:

Combinations requiring precautions during use

•**Fluoroquinolones:** in case of a concomitant use of Diaglycon MR 60mg and a fluoroquinolone, the patient should be warned of the risk of dysglycaemia, and the importance of blood glucose monitoring should be emphasized.

Combination which must be taken into account

•**Anticoagulant therapy** (e.g. Warfarin ...): Sulfonylureas may lead to potentiation of anticoagulation during concurrent treatment. Adjustment of the anticoagulant may be necessary

4.6 Fertility, pregnancy and lactation

Pregnancy

There is no or limited amount of data (less than 300 pregnancy outcomes) from the use of gliclazide during in pregnant women, even though there are few data with other sulfonylurea.

As a precautionary measure, it is preferable to avoid the use of Gliclazide during pregnancy. Control of diabetes should be obtained before the time of conception to reduce the risk of congenital abnormalities linked to uncontrolled diabetes.

Oral hypoglycaemic agents are not suitable, insulin is the drug of first choice for treatment of diabetes during pregnancy. It is recommended that oral hypoglycaemic therapy is changed to insulin before a pregnancy is attempted, or as soon as pregnancy is discovered.

Breast-feeding

It is unknown whether gliclazide or its metabolites are excreted in human milk. Given the risk of neonatal hypoglycaemia, the product is therefore contra-indicated in breast-feeding mother. A risk to the newborns/infants cannot be excluded.

Fertility

No fertility data is available for humans.

4.7 Effects on ability to drive and use machines

Diaglycon MR has no or negligible influence on the ability to drive and use machines. However, patients should be made aware of the symptoms of hypoglycaemia and should be careful if driving or operating machinery, especially at the beginning of treatment.

4.8 Undesirable effects

Based on the experience with gliclazide, the following undesirable effects have been reported.

The most frequent adverse reaction with gliclazide is hypoglycaemia.

As for other sulfonylureas, treatment with Diaglycon MR can cause hypoglycaemia, if mealtimes are irregular and, in particular, if meals are skipped. Possible symptoms of hypoglycaemia are: headache, intense hunger, nausea, vomiting, lassitude, sleep disorders, agitation, aggression, poor concentration, reduced awareness and slowed reactions, depression, confusion, visual and speech disorders, aphasia, tremor, paresis, sensory disorders, dizziness, feeling of powerlessness, loss of self-control, delirium, convulsions, shallow respiration, bradycardia, drowsiness and loss of consciousness, possibly resulting in coma and lethal outcome.

In addition, signs of adrenergic counter-regulation may be observed: sweating, clammy skin, anxiety, tachycardia, hypertension, palpitations, angina pectoris and

cardiac arrhythmia. Usually, symptoms disappear after intake of carbohydrates (sugar). However, artificial sweeteners have no effect. If a hypoglycaemic episode is severe or prolonged, and even if it is temporarily controlled by intake of sugar, immediate medical treatment or even hospitalisation is required.

Other undesirable effects:

Gastrointestinal disturbances, including abdominal pain, nausea, vomiting dyspepsia, diarrhoea, and constipation have been reported: if these should occur they can be avoided or minimised if gliclazide is taken with breakfast.

The following undesirable effects have been more rarely reported:

•**Skin and subcutaneous tissue disorders:** rash, pruritus, urticaria, angioedema, erythema, maculopapular rashes, bullous reactions (such as Stevens-Johnson syndrome and toxic epidermal necrolysis and autoimmune bullous disorders), and exceptionally, drug rash with eosinophilia and systemic symptoms (DRESS).

•**Blood and lymphatic system disorders:** Changes in haematology are rare. They may include anaemia, leucopenia, thrombocytopenia, granulocytopenia. These are in general reversible upon discontinuation of medication.

•**Hepato-biliary disorders:** raised hepatic enzyme levels (AST, ALT, alkaline phosphatase), hepatitis (isolated reports). Discontinue treatment if cholestatic jaundice appears. These symptoms usually disappear after discontinuation of treatment.

•**Eye disorders:** transient visual disturbances may occur especially on initiation of treatment, due to changes in blood glucose levels.

Class attribution effects:

As for other sulfonylureas, the following adverse events have been observed: cases of erythrocytopenia, agranulocytosis, haemolytic anaemia, pancytopenia, allergic vasculitis, hyponatraemia, elevated liver enzyme levels and even impairment of liver function (e.g. with cholestasis and jaundice) and hepatitis which regressed after withdrawal of the sulfonylurea or led to life-threatening liver failure in isolated cases.

4.9 Overdose

An overdose of sulfonylurea may cause hypoglycaemia.

Moderate symptoms of hypoglycaemia, without any loss of consciousness or neurological signs, must be corrected by carbohydrate intake, dose adjustment and/or change of diet. Strict monitoring should be continued until the doctor is sure that the patient is out of danger.

Severe hypoglycaemic reactions, with coma, convulsions or other neurological disorders are possible and must be treated as a medical emergency, requiring immediate hospitalisation.

If hypoglycaemic coma is diagnosed or suspected, the patient should be given a rapid I.V. injection of 50 mL of concentrated glucose solution (20 to 30 %). This should be followed by continuous infusion of a more dilute glucose solution (10 %) at a rate that will maintain blood glucose levels above 1 g/L. Patients should be monitored closely and, depending on the patient's condition after this time, the doctor will decide if further monitoring is necessary.

Dialysis is of no benefit to patients due to the strong binding of gliclazide to proteins.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: sulfonamides, urea derivatives. ATC code: A10BB09

Mechanism of action

Gliclazide is a hypoglycaemic sulfonylurea oral antidiabetic active substance differing from other related compounds by an N-containing heterocyclic ring with an endocyclic bond.

Gliclazide reduces blood glucose levels by stimulating insulin secretion from the β -cells of the islets of Langerhans. Increase in postprandial insulin and C-peptide secretion persists after two years of treatment. In addition to these metabolic properties, gliclazide has haemovascular properties.

Pharmacodynamics effects

Effects on insulin release:

In type 2 diabetics, gliclazide restores the first peak of insulin secretion in response to glucose and increases the second phase of insulin secretion. A significant increase in insulin response is seen in response to stimulation induced by a meal or glucose.

Haemovascular properties:

Gliclazide decreases microthrombosis by two mechanisms which may be involved in complications of diabetes:

•A partial inhibition of platelet aggregation and adhesion, with a decrease in the markers of platelet activation (beta thromboglobulin, thromboxane B2).

•An action on the vascular endothelium fibrinolytic activity with an increase in tPA activity.

5.2 Pharmacokinetic properties

Absorption

Plasma levels increase progressively during the first 6 hours, reaching a plateau which is maintained from the sixth to the twelfth hour after administration. Intra-individual variability is low. Gliclazide is completely absorbed. Food intake does not affect the rate or degree of absorption.

Distribution

Plasma protein binding is approximately 95%. The volume of distribution is around 30 litres. A single daily intake of Diaglycon maintains effective gliclazide plasma concentrations over 24 hours.

Biotransformation

Gliclazide is mainly metabolised in the liver and excreted in the urine: less than 1% of the unchanged form is found in the urine. No active metabolites have been detected in plasma.

Elimination

The elimination half-life of gliclazide varies between 12 and 20 hours.

Linearity/non-linearity

The relationship between the dose administered ranging up to 120mg and the area under the concentration time curve is linear.

Special populations

Elderly

No clinically significant changes in pharmacokinetic parameters have been observed in elderly patients.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Lactose Monohydrate, Maltodextrin, Hypromellose, Magnesium Stearate, Colloidal Silicon Dioxide

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

Please refer to the expiry date on the product labels.

6.4 Special precautions for storage

Store below 30°C.

6.5 Nature and contents of container

Diaglycon MR 30mg Modified Release Tablets

Clear PVC/PE/PVDC/Alu blisters in carton of 1 x 10's, 3 x 10's, 6 x 10's, 10 x 10's, 50 x 10's, 100 x 10's, 1 x 14's, 2 x 14's, 4 x 14's, 6 x 14's, 8 x 14's, 12 x 14's. Not all pack sizes will be marketed.

Diaglycon MR 60mg Modified Release Tablets

Clear PVC/PE/PVDC/Alu blister in carton of 1 x 15's, 2 x 15's, 4 x 15's, 6 x 15's, 8 x 15's, 12 x 15's, 1 x 10's, 3 x 10's, 6 x 10's, 10 x 10's, 50 x 10's, 100 x 10's. Not all pack sizes will be marketed.

6.6 Special precautions for disposal

No special requirements

7. MANUFACTURER

Manufactured by:

Novugen Pharma Sdn. Bhd.
No.27, Jalan Lengkok Teknologi 2,
Taman Teknologi Enstek Fasa 1,
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Product Registration Holder:

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8. DATE OF REVISION

24/08/2022

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